Algebra 1: Polynomials

Foundation Level

Name: ____________________
Date: ____________________

Question 1: Which of the following expressions is a polynomial?

A) $\frac{3}{x} + 5$
B) $2x^2 - 4x + 7$
C) $\sqrt{x} - 10$
D) $4x^{-2} + 3x$
E) $2^x + 5$

Question 2: Simplify the expression: $(3x^2 + 2x - 5) + (x^2 - 5x + 2)$.

A) $4x^2 - 3x - 3$
B) $4x^2 + 7x + 7$
C) $3x^4 - 3x^2 - 3$
D) $4x^2 - 3x + 3$
E) $2x^2 - 3x - 7$

Question 3: What is the degree of the polynomial $7x^4 - 2x^3 + 5x - 1$?

A) 1
B) 2
C) 3
D) 4
E) 7

Question 4: Subtract $(2x^2 - 4)$ from $(5x^2 + 6)$.

A) $3x^2 + 2$
B) $7x^2 + 10$
C) $3x^2 + 10$
D) $3x^2 - 10$
E) $-3x^2 - 10$

Question 5: Multiply: $4x(x^2 - 3x + 2)$.

A) $4x^3 - 12x^2 + 8x$
B) $4x^2 - 12x + 8$
C) $4x^3 - 3x + 2$
D) $x^2 - 7x + 8$
E) $4x^3 - 12x^2 + 6x$

Question 6: Find the product: $(x+3)(x+5)$.

A) x^2+15
B) x^2+8x+8
C) $x^2+15x+8$
D) $2x+8$
E) $x^2+8x+15$

Question 7: Simplify the following expression using the difference of squares: $(2x-3)(2x+3)$.

A) $4x^2+9$
B) $4x^2-6$
C) $4x^2-9$
D) $2x^2-3$
E) $4x^2-12x+9$

Question 8: Evaluate the polynomial $P(x)=x^2-4x+1$ when $x=3$.

A) 2
B) -2
C) 4
D) -4
E) 0

Question 9: Simplify the expression by distributing and combining like terms. Show all your work: $3(x^2-2x+4)-2(2x^2+x-5)$.

Question 10: A rectangle has a length represented by the expression $(x+6)$ and a width represented by $(x-2)$. Write a polynomial in standard form that represents the area of the rectangle. Show your steps.

Chef's Tips (Hints):

- When adding or subtracting polynomials, only combine terms that have the same variable and the same exponent (like terms).
- Remember that the degree of a polynomial is determined by the highest exponent of the variable in the expression.
- When subtracting a polynomial, be sure to distribute the negative sign to every term inside the parentheses.